

2.1 Equations of Motion

Question Paper

Course	CIEA Level Physics
Section	2. Kinematics
Topic	2.1 Equations of Motion
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A body having uniform acceleration a increases its velocity from u to v in time t .

Which expression would **not** give a correct value for the body's displacement during time t ?

A. $ut + \frac{1}{2}at^2$

B. $vt - \frac{1}{2}at^2$

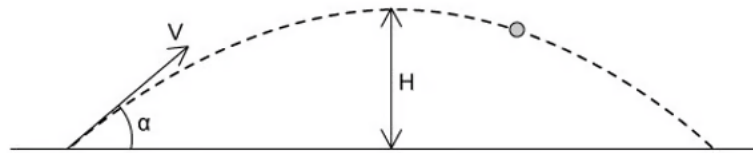
C. $\frac{(v+u)(v-u)}{2a}$

D. $\frac{(v-u)t}{2}$

[1 mark]

Question 2

A cannon fires a cannonball with an initial speed v at an angle α to the horizontal.



Which equation is correct for the maximum height H reached?

A. $H = \frac{v \sin \alpha}{2g}$

B. $H = \frac{g \sin \alpha}{2v}$

C. $H = \frac{(v \sin \alpha)^2}{2g}$

D. $H = \frac{g^2 \sin \alpha}{2v}$

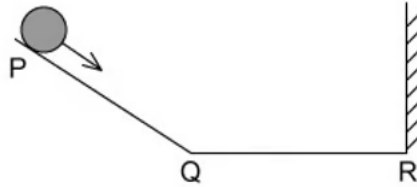
[1 mark]

Question 3

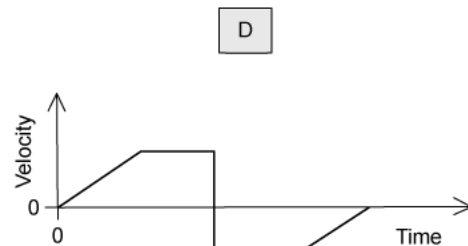
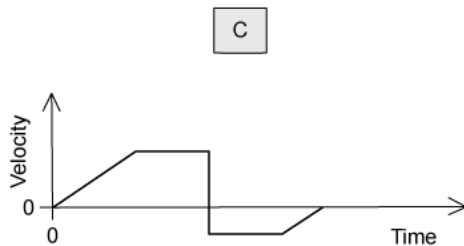
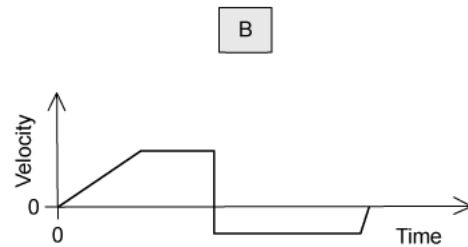
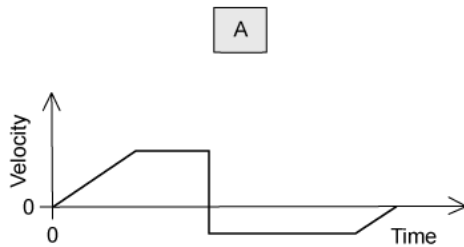
A ball is released from rest on a smooth slope PQR.

It moves down the slope, along a smooth horizontal surface QR and rebounds inelastically at R.

Then it moves back to Q and comes to rest momentarily somewhere on PQ.



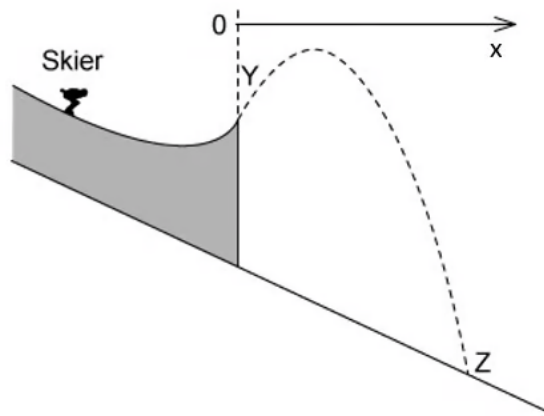
Which velocity-time graph represents the motion of the ball?



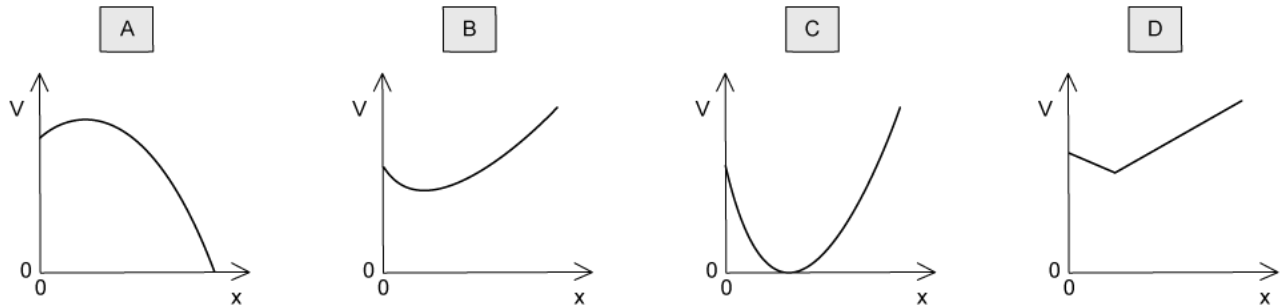
[1 mark]

Question 4

The dotted line shows the path of a competitor in a ski-jumping competition.



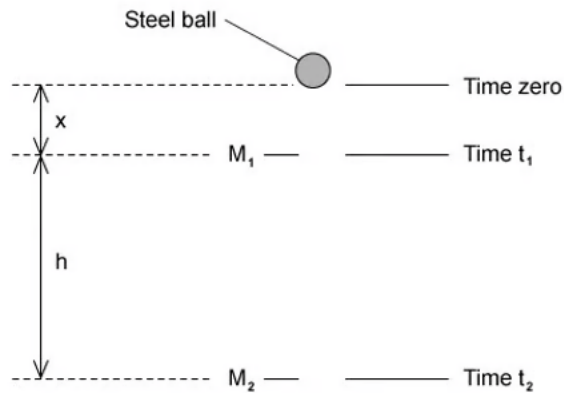
Ignoring air resistance, which graph best represents the variation of his speed v with the horizontal distance x covered from the start of his jump at Y before landing at Z?



[1 mark]

Question 5

Two markers M_1 and M_2 are set up a vertical distance h apart.



A steel ball is released at time zero from a point a distance x above M_1 . The ball reaches M_1 at time t_1 and reaches M_2 at time t_2 . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

- A. $\frac{2h}{t_2^2}$
- B. $\frac{2h}{(t_1 + t_2)}$
- C. $\frac{2h}{t_2^2 - t_1^2}$
- D. $\frac{2h}{(t_2 - t_1)^2}$

[1 mark]

Question 6

A sprinter runs a 200m race in a straight line. He accelerates from the starting block at a constant acceleration of 2.5 m s^{-2} to reach his maximum speed of 10 m s^{-1} . He maintains this speed until he crosses the finish line.

Which time does it take the sprinter to run the race?

- A. 8 s
- B. 20 s
- C. 22 s
- D. 40 s

[1 mark]

Question 7

In order that a train can stop safely, it will always pass a signal showing a yellow light before it reaches a signal showing a red light.

Drivers apply the brake at the yellow light and this results in a uniform deceleration to stop exactly at the red light. The distance between the red and yellow lights is x .

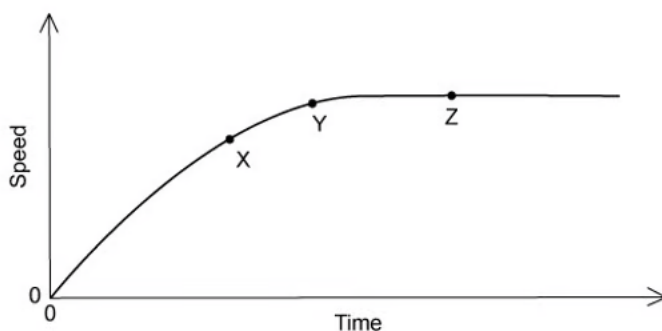
What must be the minimum distance between the lights if the train speed is increased by 25 %, without changing the deceleration of the trains?

- A. $1.20x$
- B. $1.25x$
- C. $1.44x$
- D. $1.56x$

[1 mark]

Question 8

A raindrop falls vertically from rest in air. The variation with time of the speed of the raindrop is shown in the graph.



Which statement about the raindrop is correct?

- A. at point X, the raindrop has an acceleration of 9.81 m s^{-2}
- B. at point Z, the force on the raindrop due to air resistance has reached its maximum value and so the acceleration of the raindrop has also reached its maximum value
- C. at point Z, the force due to air resistance is equal and opposite to the weight of the raindrop and so the speed of the raindrop is zero
- D. the resultant force on the raindrop at point Y is less than the resultant force on the raindrop at point X

[1 mark]

Question 9

An aeroplane travels at an average speed of 700 km h^{-1} on an outward flight and at 300 km h^{-1} on the return flight over the same distance.

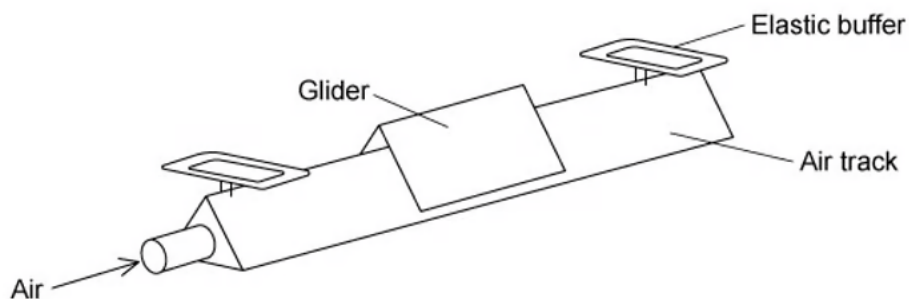
What is the average speed of the whole flight?

- A. 400 km h^{-1}
- B. 420 km h^{-1}
- C. 480 km h^{-1}
- D. 500 km h^{-1}

[1 mark]

Question 10

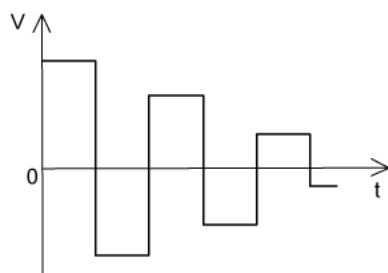
A small glider moves along a friction-free horizontal air track as shown below.



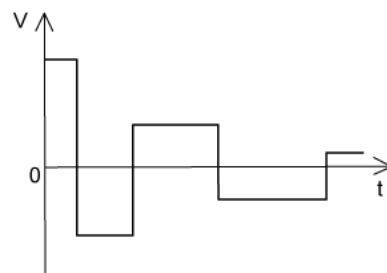
At each end of the air track there is a perfectly elastic buffer.

Which graph represents the variation with time t of the velocity v of the glider as it moves between the two buffers?

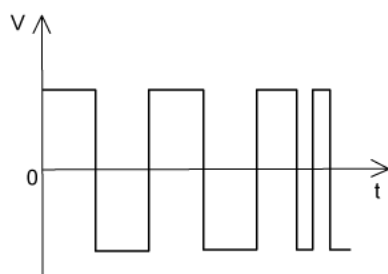
A



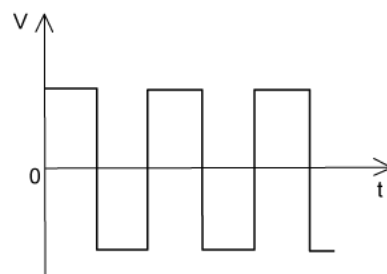
B



C



D



[1 mark]

Model Answers: Hard

1

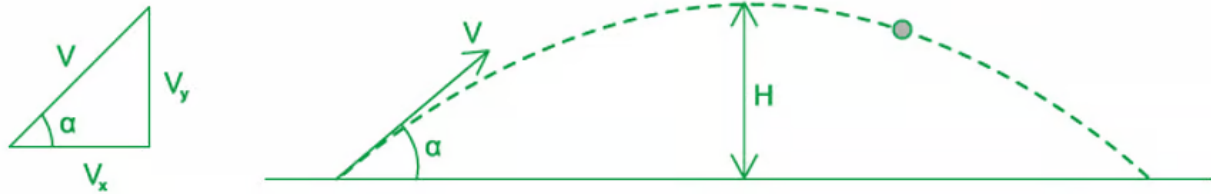
The correct answer is **D** because:

- *Average speed* = $\frac{v+u}{2}$
- *Distance* = *speed* $\times$ *time* = $\frac{(v+u)t}{2}$
- D would be correct if it had a plus sign

A is incorrect as	this equation is equal to displacement $s = ut + \frac{1}{2}at^2$
B is incorrect as	if we combine the equations $s = ut + \frac{1}{2}at^2$ and $v = u + at$ $u = v - at$ $s = (v - at)t + \frac{1}{2}at^2$ $s = vt - at^2 + \frac{1}{2}at^2$ $s = vt - \frac{1}{2}at^2$
C is incorrect as	if we take the equation $v^2 = u^2 + 2as$ $2as = v^2 - u^2$ $s = \frac{v^2 - u^2}{2a}$ we can factorise $(v^2 - u^2) = (v + u)(v - u)$ so, $\frac{(v + u)(v - u)}{2a}$

2

The correct answer is **C** because:



$$\sin \alpha = \frac{V_y}{V}$$

$$\cos \alpha = \frac{V_x}{V}$$

Vertical component of velocity

Horizontal component of velocity

$$\uparrow V_y = V \sin \alpha$$

$$\rightarrow V_x = V \cos \alpha$$

$$\uparrow u = V \sin \alpha \quad V = 0 \quad a = -g \quad S = H$$

List the quantities that are known

$$V^2 = u^2 + 2as$$

Find out which equation of motion contains these quantities

$$0 = (V \sin \alpha)^2 - 2gH$$

Insert the quantities into the equation

Add 2gH to both sides

$$2gH = (V \sin \alpha)^2$$

Divide both sides by 2g

$$H = \frac{(V \sin \alpha)^2}{2g}$$

3

The correct answer is **A** because:

- From P to Q, the weight of the ball causes it to accelerate uniformly. Therefore, the

increase in speed is uniform so the gradient is constant

- From Q to R, the motion of the ball is horizontal, so the acceleration of free fall does not affect its motion which is shown on the graph as a horizontal line
- After the collision with the wall, the kinetic energy decreases since the collision is inelastic, this results in a decrease in speed ($KE = \frac{1}{2} mv^2$)
- Therefore, the ball will take longer to return to the slope, meaning the negative horizontal velocity should be longer

B & D are incorrect because the negative horizontal velocities on these graphs are too short, and the value of the negative velocity in graph D has not decreased

C is incorrect because the gradient of the graph in the final section is too steep, since the gradient represents the acceleration due to gravity, it should be the same as the gradient in the first section

4

The correct answer is **B** because:

- At the skier's maximum height, the speed is at a minimum since gravitational potential is at a maximum and kinetic energy is at a minimum
- The speed does not reach zero since it still has horizontal velocity
- Just before landing, the speed is not zero. Since Y is higher than Z, the person has both potential energy and kinetic energy at Y
- Because of the conservation of energy, kinetic energy is higher at Z so the speed is higher here

A is incorrect because this graph shows the skier has maximum kinetic energy (highest v) and minimum gravitational energy at the top of the path, which is the opposite to what really happens

C is incorrect because the speed doesn't reach zero as the skier still has horizontal velocity at the top of the path

D is incorrect because the change in v and x is not uniform due to the acceleration due to gravity

5

The correct answer is **C** because:

- From the origin to marker M_1 :
- Initially the ball is at rest, so:
 - $u = 0 \text{ m s}^{-1}$
 - $t = t_1$
 - $s = x$
- The equation that links s , u , t and a is
 - $s = ut + \frac{1}{2}at^2$
 - $x = 0 + \frac{1}{2}at_1^2$
 - $x = \frac{1}{2}at_1^2 \quad (1)$
- From the origin to marker M_2 :
 - $u = 0 \text{ m s}^{-1}$
 - $t = t_2$
 - $s = x + h$
- Using the same equation as before:
 - $s = ut + \frac{1}{2}at^2$
 - $x + h = 0 + \frac{1}{2}at_2^2$
 - $x + h = \frac{1}{2}at_2^2 \quad (2)$
- If we take equation (1) from (2):
 - $x + h - x = \frac{1}{2}at_2^2 - \frac{1}{2}at_1^2$
 - $h = \frac{1}{2}a(t_2^2 - t_1^2)$
 - $a = \frac{2h}{(t_2^2 - t_1^2)}$

6

The correct answer is **C** because:

- There are two parts to this motion
 - Motion with constant acceleration of 2.5 m s^{-2}
 - Motion with constant speed of 10 m s^{-1}

- $x + h - x = \frac{1}{2}at_2^2 - \frac{1}{2}at_1^2$
- $h = \frac{1}{2}a(t_2^2 - t_1^2)$
- $a = \frac{2h}{(t_2^2 - t_1^2)}$

6

The correct answer is **C** because:

- There are two parts to this motion
 - Motion with constant acceleration of 2.5 m s^{-2}
 - Motion with constant speed of 10 m s^{-1}
- If the duration of the acceleration is t_1 then we want to use an equation to find this, where; $t = t_1$, $a = 2.5 \text{ m s}^{-2}$, $u = 0 \text{ m s}^{-1}$, $v = 10 \text{ m s}^{-1}$
 - $v = u + at$
 - $v - u = at_1$
 - $t_1 = \frac{v - u}{a} = \frac{10}{2.5} = 4 \text{ s}$
- If the distance travelled during the acceleration is s_1 then we want to use a different equation to find this
 - $v^2 = u^2 + 2as$
 - $v^2 - u^2 = 2as_1$
 - $s_1 = \frac{v^2 - u^2}{2a} = \frac{10^2}{2 \times 2.5} = 20 \text{ m}$
- The remainder of the 200m race is run at a constant speed of 10 m s^{-1} , so the remaining distance is equal to $200 - 20 = 180 \text{ m}$
- If the time taken to cover the remaining distance is t_2 , then we can use
 - $\text{speed} = \frac{\text{distance}}{\text{time}}$
 - $t_2 = \frac{180}{10} = 18 \text{ s}$
- Total time = $t_1 + t_2 = 4 + 18 = 22 \text{ s}$

7

The correct answer is **D** because:

- Let's consider the quantities we have been given
 - The initial speed of the train is u
 - The distance between the red and yellow lights is x
 - Final speed of train = 0
- The equation that links u , s , v and a is
 - $v^2 = u^2 + 2as$
 - $0 = u^2 + 2ax$
 - $u^2 \propto x$
- If the initial speed increases by 25 % : 100 % + 25 % = 125 %
- Therefore, new speed $u_{\text{new}} = 1.25u$
- So, new distance $x_{\text{new}} \propto u_{\text{new}}^2 = (1.25u)^2 = 1.56u^2 = 1.56x$

8

The correct answer is **D** because:

- The gradient of a speed-time graph is acceleration
- Between points X and Z the gradient is decreasing so the acceleration is decreasing until the raindrop reaches a steady speed
- Force is directly proportional to acceleration
- The magnitude of the acceleration is lower at Y is less than at X, therefore, the magnitude of the resultant force is lower at Y too

A is incorrect because the magnitude of air resistance is beginning to increase around point X so the acceleration is less than 9.81 m s^{-2}

B is incorrect because at point Z, it is correct that the force on the raindrop due to air resistance has reached its maximum value but this means the acceleration of the raindrop has reached its **minimum** value as the resultant force is at a minimum

C is incorrect because at point Z, it is correct that the force due to air resistance is equal and opposite to the weight of the raindrop but the speed of the raindrop is **not** zero, it has reached a steady speed and is no longer accelerating

9

The correct answer is **B** because:

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$$

- $\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$
- Let distance travelled for outward flight, which is equal to the distance travelled for inward flight = d
- So, the total distance travelled = $2d$
- For each flight speed, the time will be different since the speed is different for the flights
- Let time for outward flight = $t_{out} = \frac{d}{700}$
- Let time for inward flight = $t_{in} = \frac{d}{300}$
- Total time, $t = t_{out} + t_{in} = \frac{d}{300} + \frac{d}{700} = \frac{d}{210}$
- $\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$
- $\text{Average speed} = \frac{2d}{d/210} = 420 \text{ km h}^{-1}$

10

The correct answer is **D** because:

- A friction-free horizontal air track means that no work needs to be done against friction, therefore, the kinetic energy does not decrease
- A perfectly elastic buffer means the collision of the glider is perfectly elastic, therefore, both momentum and energy are conserved
- This means the magnitude of the velocity should not decrease, so the height of the peaks on the graph should be the same
- Since the speed remains constant, the time between each collision should be the same, so the gaps between each peak should be the same

A & B are incorrect because these graphs show the magnitude of the velocity changes as the peaks are different sizes

C is incorrect because this graph shows the time between collisions varies as the gaps between some of the peaks are not the same



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